Uncontrolled document exported 24.09.2026
View online at cpg.stjohn.org.nz

Sodium Bicarbonate 8.4%

Mechanism of action

- 8.4% sodium bicarbonate is a systemic alkalinising agent. It increases plasma bicarbonate, buffers hydrogen ions and raises the blood pH.
- Following crush injury 8.4% sodium bicarbonate has a role because:
  - Sodium ions help protect cardiac cell membranes from the effects of hyperkalaemia.
  - A rise in pH results in potassium moving into the cells, which lowers the potassium concentration in blood.
  - A rise in urinary pH reduces myoglobin deposition in the kidneys.

Authorised practice levels

- ICPs and CCPs.

Indications

- ✓ Release syndrome following crush injury in an adult.
- ✓ Known or suspected hyperkalaemia with severe ECG changes.
- ✓ Cardiac arrest secondary to hyperkalaemia.
- ✓ Suspected cyclic antidepressant poisoning with QRS prolongation.
- ✓ Severe TBI and intubated with clinical signs of raised intracranial pressure.

Contraindications

- ✗ Known severe allergy.

Cautions

- ⚠ IV access via a small vein. 8.4% sodium bicarbonate is hyperosmolar and will cause venous irritation if administered via a small vein.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated, but 8.4% sodium bicarbonate should be administered if indicated.

Sodium Bicarbonate 8.4%

Dosage

- Cyclic antidepressant poisoning:
  - 100 ml IV for an adult.
  - 2 ml/kg for a child.
- Severe TBI, intubated and clinical signs of raised intracranial pressure:
  - 100 ml IV for an adult.
  - 2 ml/kg for a child.
- Hyperkalaemia with severe ECG changes:
  - 100 ml IV for an adult.
  - Repeat the dose after 20 minutes if severe ECG changes persist or recur.
  - Seek clinical advice for a child.
- Cardiac arrest secondary to hyperkalaemia:
  - 100 ml IV for an adult.
  - Repeat the dose if ROSC is not achieved within 10 minutes of the first dose.
  - Seek clinical advice for a child.
- Release syndrome following crush injury:
  - 100 ml IV for an adult.
  - Administer further doses if signs of hyperkalaemia (including dysrhythmia) occur.
  - Seek clinical advice for a child.

Administration

- Cardiac arrest secondary to hyperkalaemia: Administer IV as a bolus, via a running line.
- All other indications: Administer IV over approximately one minute via a running line.
- Do not mix with other medicines as precipitation will occur. If other medicines are being administered via the same vein, ensure a minimum IV flush of 50 ml of 0.9% sodium chloride between medicines.

Common adverse effects

- None.

Sodium Bicarbonate 8.4%

Usual onset of effect

- 1-2 minutes.

Usual duration of effect

- 1-2 hours.

Usual preparation

- Ampoule containing 1 mmol/ml of sodium bicarbonate.
- Ampoules may be 10 ml, 50 ml or 100 ml.

Pharmacokinetics

- 8.4% sodium bicarbonate dissociates into sodium and bicarbonate ions which are excreted in urine.
- There are no significant effects from renal impairment on administration.

Common interactions

- None.

Additional information

- Document the dose administered as mmol.

More Information

Version history

28 October 2024 1.0.5
Update: Indications and dosage updated.

11 September 2023 1.0.4
Update: New indications have been added. Dosage and administration information updated.